Santhoshi Kuppa Srinivasan

Senior Software Engineer

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https://santhoshi-k-s.github.io/portfolio-santhoshi/

Summary

Software engineer and data scientist with about five years of experience building AI-driven platforms, enterprise privacy systems, and CPQ applications at Tempero.tech, PayPal, and Configit. Experience spans agentic RAG pipelines, distributed AWS simulation infrastructure, and full-stack engineering and holds a B.E. in Computer Science with First Class Distinction

Education

B.E. Computer Science — First Class with Distinction, 9.4 CGPA (94%) · Thiagarajar College of Engineering, Tamil Nadu, India

- Class XII: 98.5%, Class X: 97%

Experience

Senior Data Scientist, Tempero.tech (2025-02 - Present)

- **Co-owned CARE platform architecture**, designing an AI-driven system translating autonomous-vehicle simulation behavior into insurer-relevant risk scores.
- **Built agentic scenario-generation pipeline**, using RAG and structured validation to convert crash descriptions into OpenSCENARIO/OpenDRIVE artefacts, cutting authoring effort ~60-70% and improving coverage ~40%.
- **Architected distributed AWS simulation pipeline**, leveraging EC2, S3, Lambda, and Batch for large-scale parallel experimentation, improving compute efficiency by ~40%.
- **Developed automated LLM quality judge**, evaluating generated scenarios for validity, realism, and standards compliance to reduce manual review effort.
- **Enhanced OSCFuzzer genetic-algorithm fuzzer**, uncovering rare safety-critical edge cases through large-scale scenario mutation across ADAS/AV failure modes.
- **Built Flask application layer**, powering CARE's structured risk-signal generation and stakeholder reporting, and led secure binary packaging for internal frameworks.
- **Supported client and partner engagements**, contributing to technical demonstrations and deep-dives for potential clients and enterprise partners.

Stack: Python, RAG, LLMs, AWS, Docker, Flask, SQL, OpenSCENARIO, OpenDRIVE, esmini, CARLA

Software Engineer, PayPal (2023-08 - 2025-02)

- **Cut incident response time ~50%**, leading a Datadog observability integration centralizing log monitoring, error tracking, and performance metrics.
- **Built compliance reporting dashboards**, developing scalable alerting infrastructure that surfaced anomalies and risk indicators in real time, reducing operating costs.
- **Implemented NL-to-SQL query workflows**, enabling natural-language interaction with compliance datasets to improve analytical accessibility for stakeholders.
- **Proposed confidential computing architecture**, for sensitive regulated data pipelines, collaborating cross-functionally with product, compliance, and engineering teams.

Stack: React JS, GraphQL, JavaScript, Node.js, Python, Java, Datadog, Jest, Figma

Software Engineer, Configit (2021-09 - 2023-08)

- **Built Sales Configurator application**, using high-performance React UI components for dynamic product modeling and rule-based configuration logic.
- **Designed backend configuration APIs**, handling complex pricing and configuration constraints for enterprise-scale CPQ platforms at European manufacturing clients.
- **Contributed to architecture and feature planning**, in a product-driven engineering environment using C#, JavaScript, GraphQL, RavenDB, and REST APIs.

Stack: C#, JavaScript, React JS, GraphQL, RavenDB, REST APIs, HTML, CSS

Technical Consultant (Intern), Blue Yonder (2021-01 - 2021-08)

- **Built Python automation tool**, for WMS application upgrades and database refresh workflows, reducing manual touch points by ~30%.
- **Standardized recurring maintenance processes**, increasing reliability in production-like environments while working across cross-functional Agile teams.

Stack: Python, MySQL, WMS

Skills

Programming Languages

- Python, JavaScript, C#, Java.

AI & Intelligent Systems

- Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic Workflows, Model Context Protocol (MCP), Deep Learning, Prompt Engineering.

Frameworks & Web

- React JS, Node.js, Flask, GraphQL, REST APIs, HTML, CSS.

Cloud & DevOps

- AWS (EC2, S3, Lambda, Batch), GCP, Docker.

Databases & Tools

- MySQL, RavenDB, Git, JIRA, Figma.

Certifications

Claude Code 101 · Anthropic ↗

Introduction to Model Context Protocol · Anthropic ↗

Model Context Protocol: Advanced Topics · Anthropic ↗

Claude 101 · Anthropic ↗

AI Fluency for Small Businesses · Anthropic ↗

Machine Learning with Python · Coursera ↗

Cloud Computing · Coursera ↗

Configit Certified Software Engineer · Credly ↗

Configuration Lifecycle Management · Credly ↗

Problem Solving (Intermediate) · HackerRank ↗